

AUDIO REPARATION OF GRAPHOPHONE

----- A Comparitive Analysis between two reparation methods



INTRODUCTION

Early gramophone recordings suffer from unique distortions (muffled highs, uneven lows) that standard noise removal can't fix, risking valuable history. We **built a gramophone model** to capture these flaws and **compared** two repair methods: **FIRWIN2** vs. **Dynamic Spectrum Matching (DSM)**. DSM proved best, enabling us to **create a universal repair model**. This model tells us exactly how to boost/cut frequencies, letting anyone restore old recordings—even without originals—helping **preserve audio heritage**.

METHODOLOGY



Data collection collect audios (original&recorded)

Implementing Algorithm 4 trials each

Quantitative Evaluation LSD & Cos Similarity

Model Construction universal model

OBJECTIVE

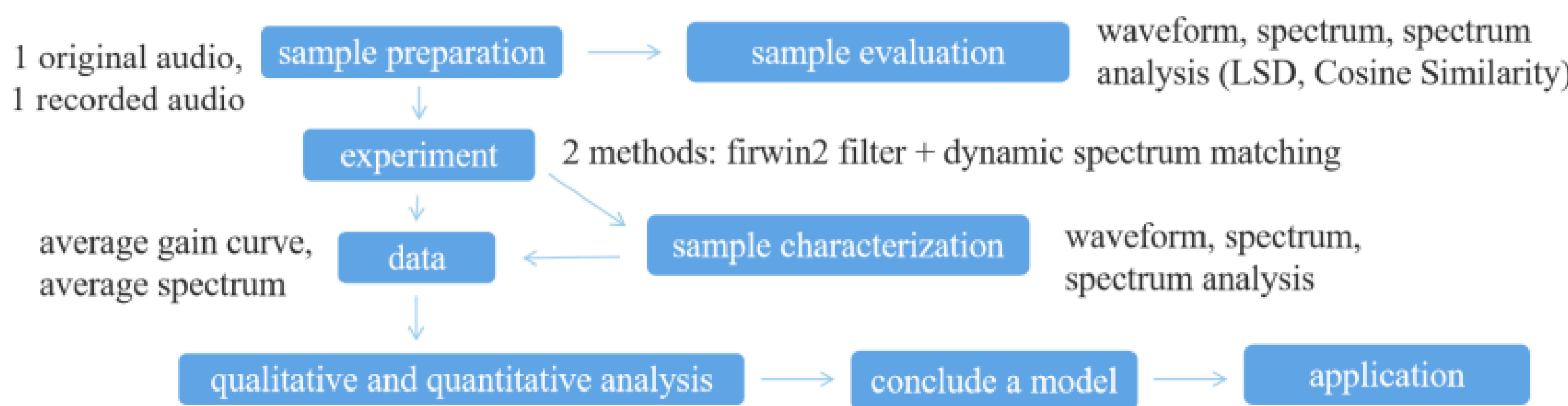
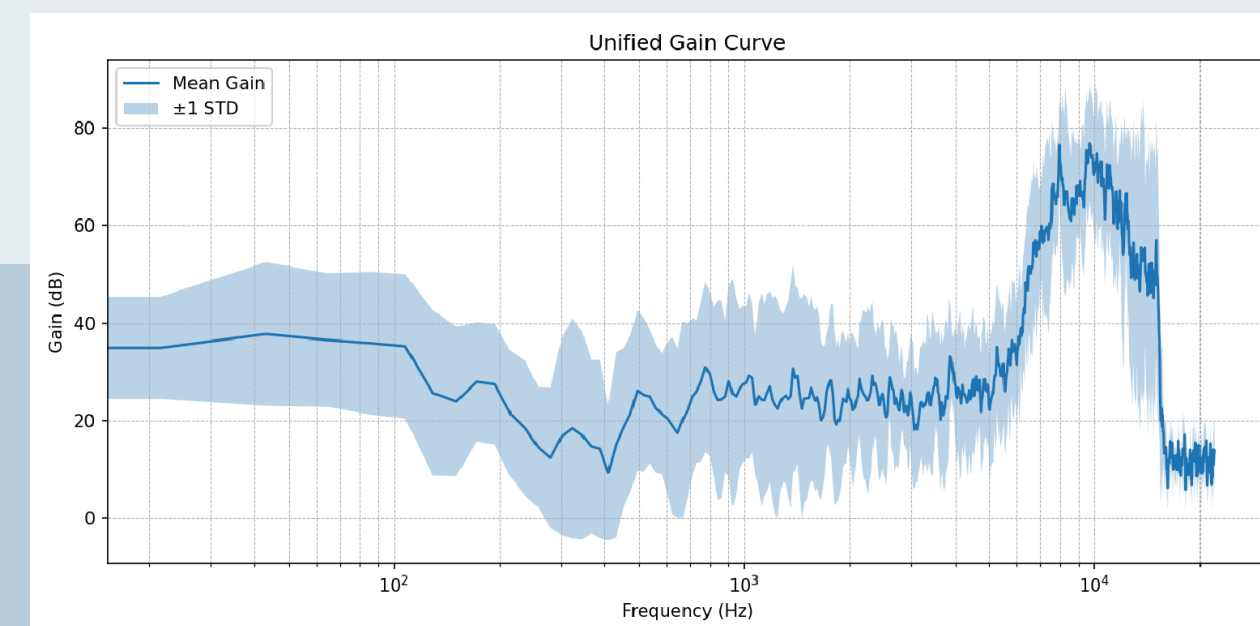
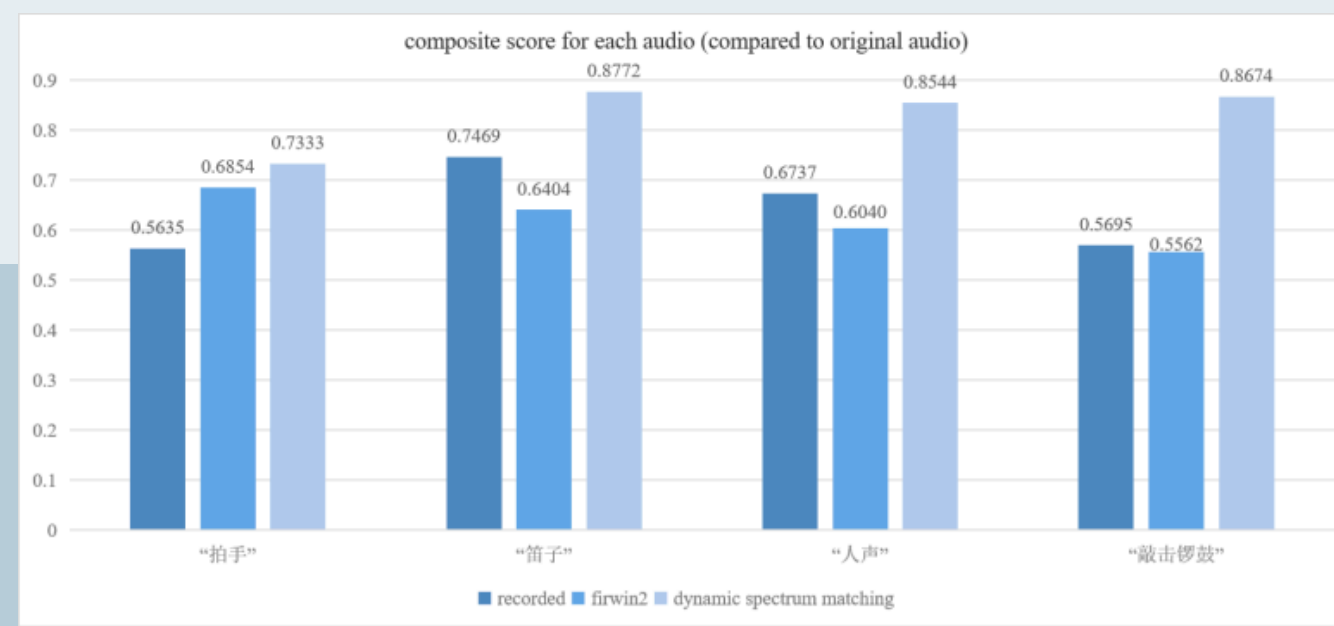
- ✓ Summarize distortion patterns of gramophone recordings
- ✓ Compare different methods of audio reparation
- ✓ Build a universal model for gramophone-recorded audio sound reparation

RESULTS

① DSM outperformed FIRWIN2: **34%** sound quality improvement

② Universal Repair **Model** Created: Generated a **frequency-specific gain curve** by averaging DSM results

③ Successfully restored historical recordings without original references

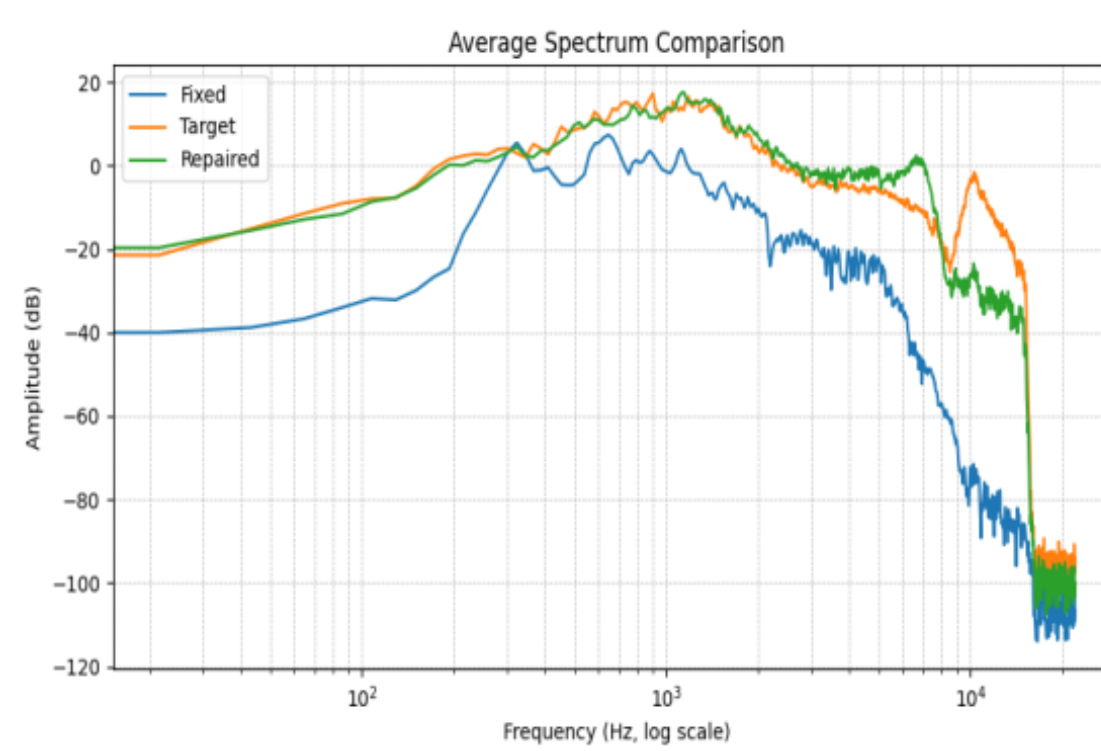


ANALYSIS

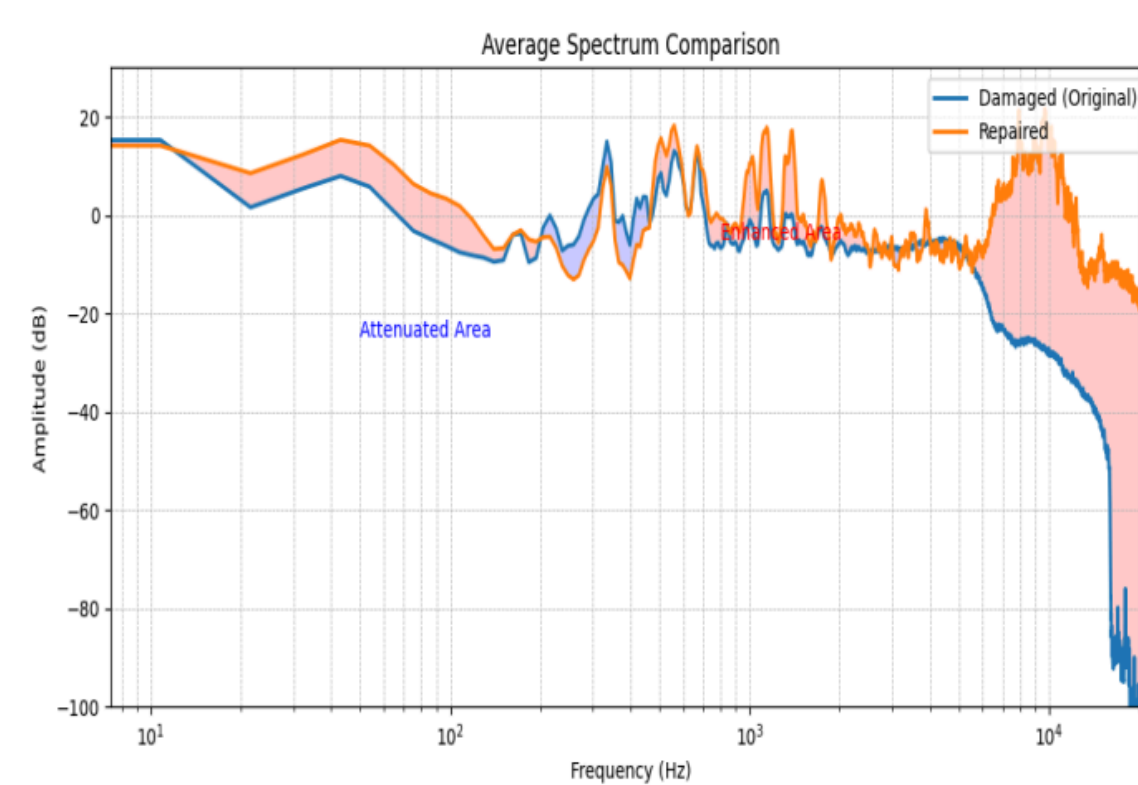
- ✓ Gramophone recordings at $>8\text{kHz}$ → muffled sound
- ✓ DSM repair beats FIRWIN2: better spectrum matching accurate timbre higher LSD & CosS score
- ✓ universal model reference-free reparation peak gain at 10^4 Hz



Spectrum Comparison between original and recorded audio



average spectrum comparison (recorded, repaired, original)



average spectrum comparison (before & after reparation using our model)

CONCLUSION

- DSM Superiority for Gramophone Repair
- Quantified Gramophone Distortion Characteristics
- Unified Frequency Compensation Model
- Effective Evaluation Framework
- Practical Solution for Preservation

Related literature

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- [10] Edmunds, H. (1888). 3rd Meeting:- "The Graphophone".
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